

Arab American University

Computer Systems Engineering Faculty

Verification and Validation

Project #1: Selenium Automated Web Testing

Submitted by:

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Part 1: Theory Introduction to Selenium

1.1 What is Selenium?

Selenium is an open-source, portable framework used for automating web browsers. It supports multiple programming languages such as Python, Java, and JavaScript, and works across all major browsers including Chrome, Firefox, and Edge. Selenium is widely used in regression testing to verify that new code changes do not break existing functionality in web applications.

Selenium operates by simulating real user interactions — clicking buttons, filling forms, navigating pages — without any manual intervention. This makes it a critical tool in modern software development and quality assurance pipelines.

1.2 Video Watched

Title: Python Selenium Tutorial #1 — Web Scraping, Bots & Testing

Channel: Tech With Tim (2.03M subscribers)

URL: <https://www.youtube.com/watch?v=Xjv1sY630Uc>

Direct Link: <https://www.youtube.com/watch?v=Xjv1sY630Uc>

The video covers how to install Selenium with Python, set up ChromeDriver, and write basic automation scripts. It demonstrates practical usage through live coding examples showing how to launch a browser, interact with elements, and verify page states.

1.3 Screenshot Watching the Video

The following screenshot was taken on Tuesday, June 9, 2026 while watching the Selenium tutorial video. The date and time are visible in the system taskbar.

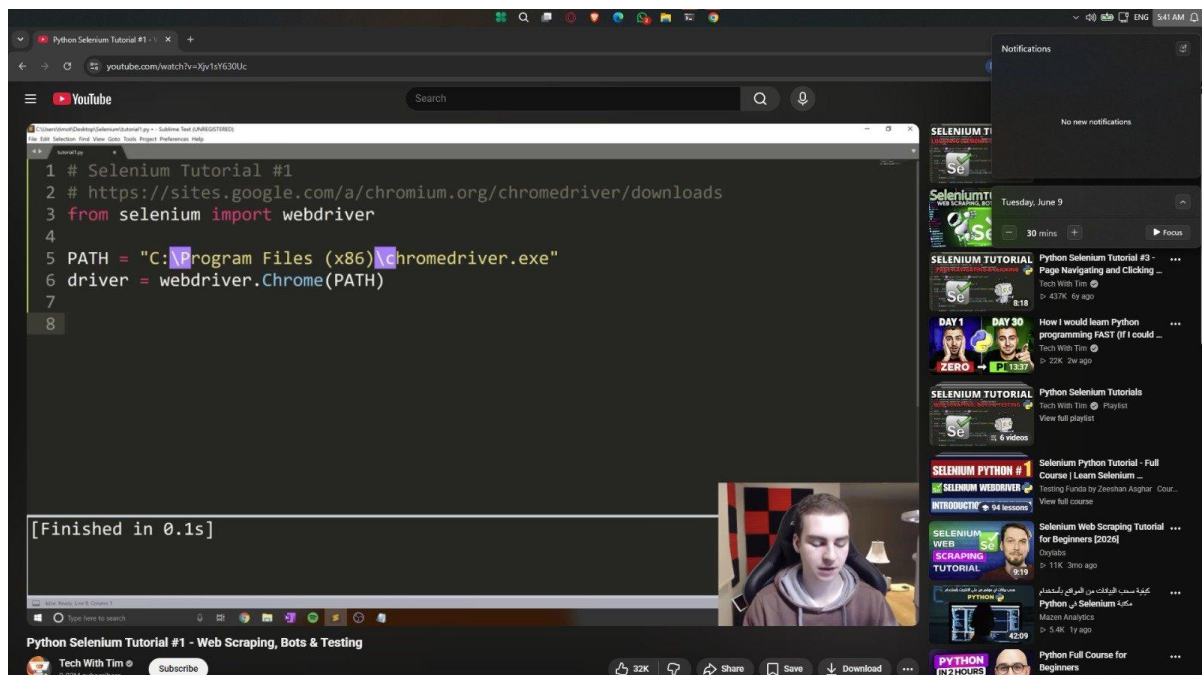


Figure 1: Screenshot taken on June 9, 2026 while watching 'Python Selenium Tutorial #1' by Tech With Tim

Part 2: Practical Installation and Testing

2.1 Installation Process

The following steps were performed to install Selenium and its dependencies on a Windows machine:

Step 1 Download and install Python 3.14.5 (64-bit) from <https://python.org>, making sure to check 'Add Python to PATH' during installation.

Step 2 Install the Selenium library using pip: `pip install selenium`

Step 3 Install webdriver-manager to automatically handle ChromeDriver downloads: `pip install webdriver-manager`

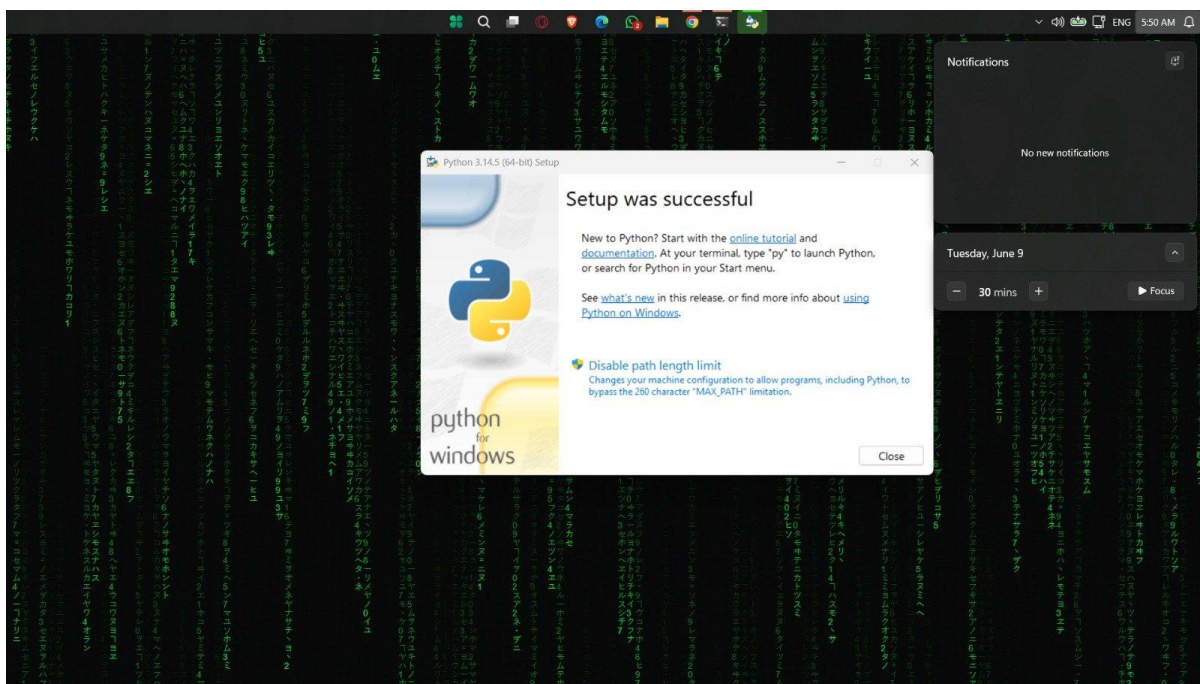


Figure 2: Python 3.14.5 setup completed successfully on June 9, 2026

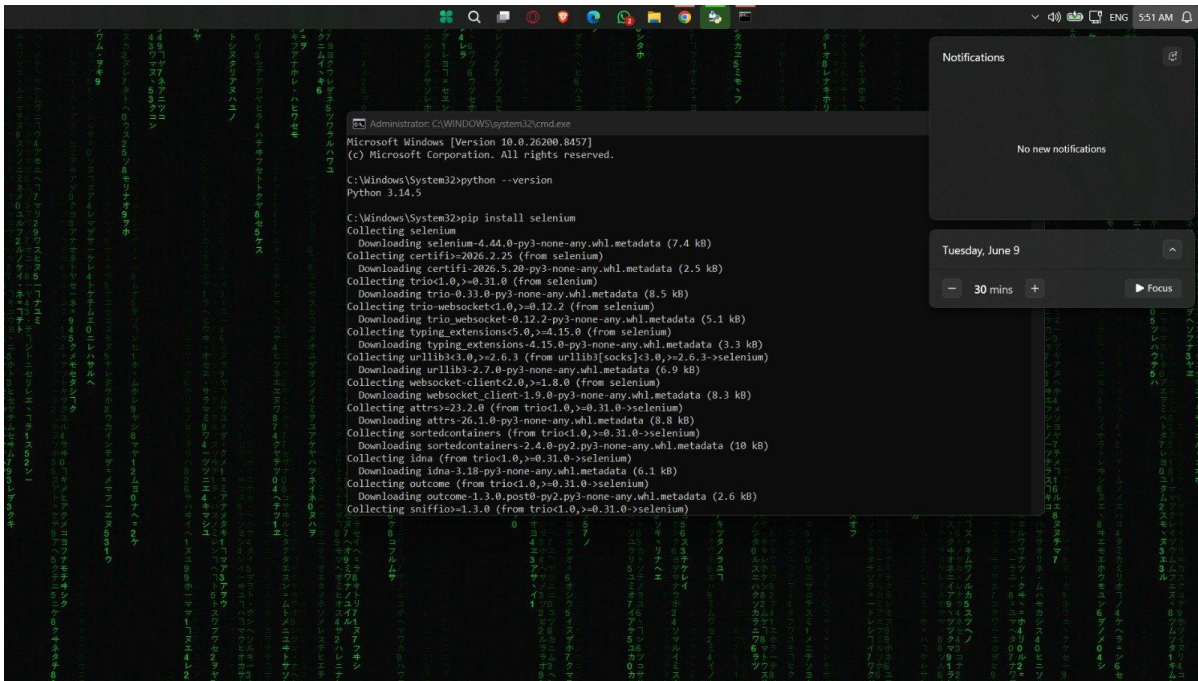


Figure 3: Installing Selenium 4.44.0 via pip — June 9, 2026

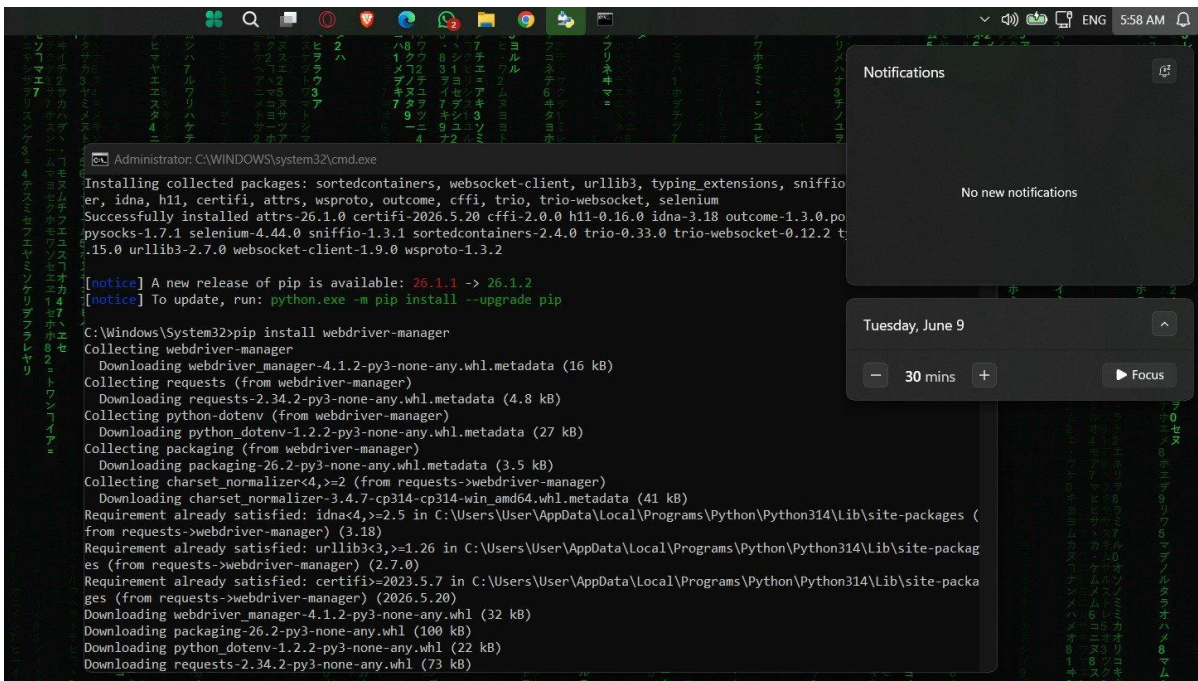


Figure 4: Installing webdriver-manager — June 9, 2026

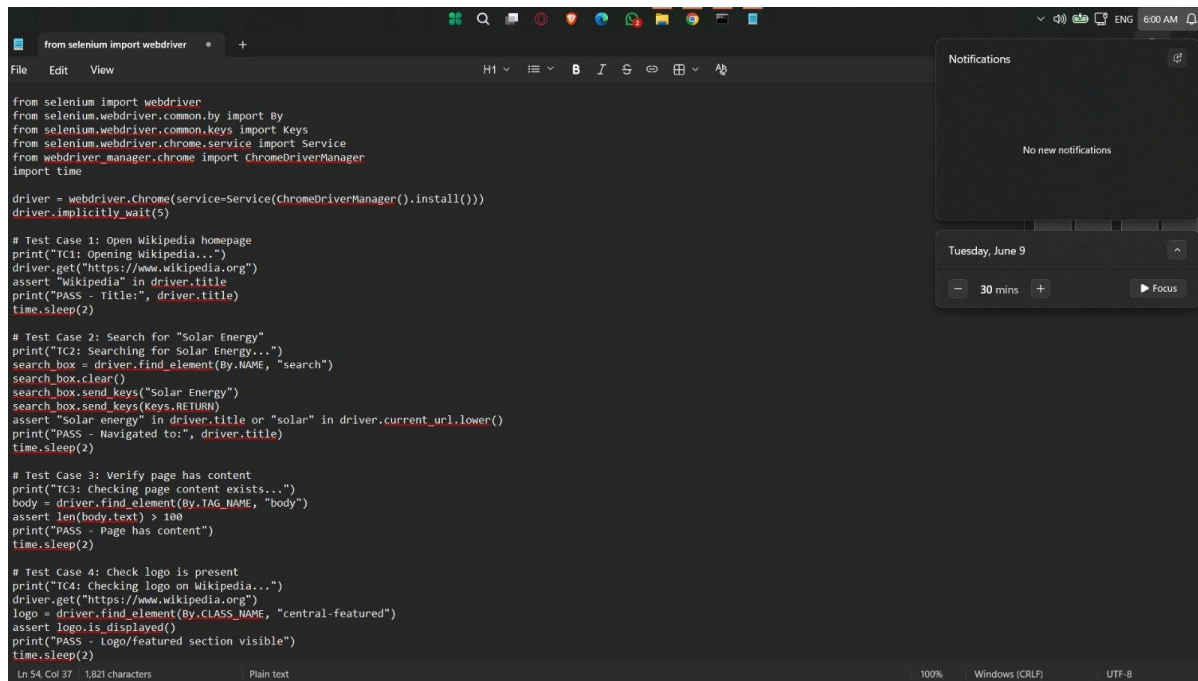
2.2 Test Cases Design

Five automated test cases were designed to test the Wikipedia website (<https://www.wikipedia.org>). Wikipedia was chosen as the target web application because it is a publicly accessible, stable site that provides a variety of testable interactions including page navigation, search functionality, and element visibility checks.

TC#	Test Case Name	Description	Expected Result	Status
TC1	Open Wikipedia Homepage	Navigate to https://www.wikipedia.org and verify the page title contains 'Wikipedia'	Title contains 'Wikipedia'	PASS
TC2	Search for Solar Energy	Locate the search box by name, type 'Solar Energy', press ENTER, and verify the URL contains 'solar'	Page navigates to Solar energy article	PASS
TC3	Verify Page Content Exists	Find the <body> element and confirm its text length exceeds 100 characters	Body text length > 100 chars	PASS
TC4	Check Homepage Featured Section	Return to Wikipedia homepage and verify the central-featured section element is displayed	Featured section is visible	PASS
TC5	Search for Palestine	From the homepage, search for 'Palestine' and verify the resulting page title contains 'Palestine'	Page navigates to Palestine article	PASS

2.3 Selenium Test Script

The following Python script was written using the Selenium WebDriver API and the webdriver-manager library. The script automatically launches a Chrome browser, executes all five test cases sequentially, and prints PASS/FAIL results to the console.



```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.chrome.service import Service
from webdriver_manager.chrome import ChromeDriverManager
import time

driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()))
driver.implicitly_wait(5)

# Test Case 1: Open Wikipedia homepage
print("TC1: Opening Wikipedia...")
driver.get("https://www.wikipedia.org")
assert "Wikipedia" in driver.title
print("PASS - Title:", driver.title)
time.sleep(2)

# Test Case 2: Search for "Solar Energy"
print("TC2: Searching for Solar Energy...")
search_box = driver.find_element(By.NAME, "search")
search_box.clear()
search_box.send_keys("Solar Energy")
search_box.send_keys(Keys.RETURN)
assert "Solar energy" in driver.title or "solar" in driver.current_url.lower()
print("PASS - Navigated to:", driver.title)
time.sleep(2)

# Test Case 3: Verify page has content
print("TC3: Checking page content exists...")
body = driver.find_element(By.TAG_NAME, "body")
assert len(body.text) > 100
print("PASS - Page has content")
time.sleep(2)

# Test Case 4: Check logo is present
print("TC4: Checking logo on wikipedia...")
driver.get("https://www.wikipedia.org")
logo = driver.find_element(By.CLASS_NAME, "central-featured")
assert logo.is_displayed()
print("PASS - Logo/featured section visible")
time.sleep(2)

# Test Case 5: Search for "Palestine" and verify result
print("TC5: Searching for Palestine...")
search_box = driver.find_element(By.NAME, "search")
search_box.clear()
search_box.send_keys("Palestine")
search_box.send_keys(Keys.RETURN)
assert "Palestine" in driver.title or "palestine" in driver.current_url.lower()
print("PASS - Navigated to:", driver.title)
time.sleep(2)

driver.quit()
print("\nAll test cases completed!")
```

Figure 5: Selenium test script written in Notepad — June 9, 2026

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.chrome.service import Service
from webdriver_manager.chrome import ChromeDriverManager
import time

driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()))
driver.implicitly_wait(5)

# Test Case 1: Open Wikipedia homepage
print("TC1: Opening Wikipedia...")
driver.get("https://www.wikipedia.org")
assert "Wikipedia" in driver.title
print("PASS - Title:", driver.title)
time.sleep(2)

# Test Case 2: Search for "Solar Energy"
print("TC2: Searching for Solar Energy...")
search_box = driver.find_element(By.NAME, "search")
search_box.clear()
search_box.send_keys("Solar Energy")
search_box.send_keys(Keys.RETURN)
assert "Solar energy" in driver.title or "solar" in driver.current_url.lower()
print("PASS - Navigated to:", driver.title)
time.sleep(2)

# Test Case 3: Verify page has content
print("TC3: Checking page content exists...")
body = driver.find_element(By.TAG_NAME, "body")
assert len(body.text) > 100
print("PASS - Page has content")
time.sleep(2)

# Test Case 4: Check logo is present
print("TC4: Checking logo on Wikipedia...")
driver.get("https://www.wikipedia.org")
logo = driver.find_element(By.CLASS_NAME, "central-featured")
assert logo.is_displayed()
print("PASS - Logo/featured section visible")
time.sleep(2)

# Test Case 5: Search for "Palestine" and verify result
print("TC5: Searching for Palestine...")
search_box = driver.find_element(By.NAME, "search")
search_box.clear()
search_box.send_keys("Palestine")
search_box.send_keys(Keys.RETURN)
assert "Palestine" in driver.title or "palestine" in driver.current_url.lower()
print("PASS - Navigated to:", driver.title)
time.sleep(2)

driver.quit()
print("\nAll test cases completed!")
```

2.4 Test Execution Results

The script was executed from the Windows Command Prompt. All five test cases passed successfully. The automated browser launched, navigated to the required pages, performed searches, and verified the expected results without any failures.

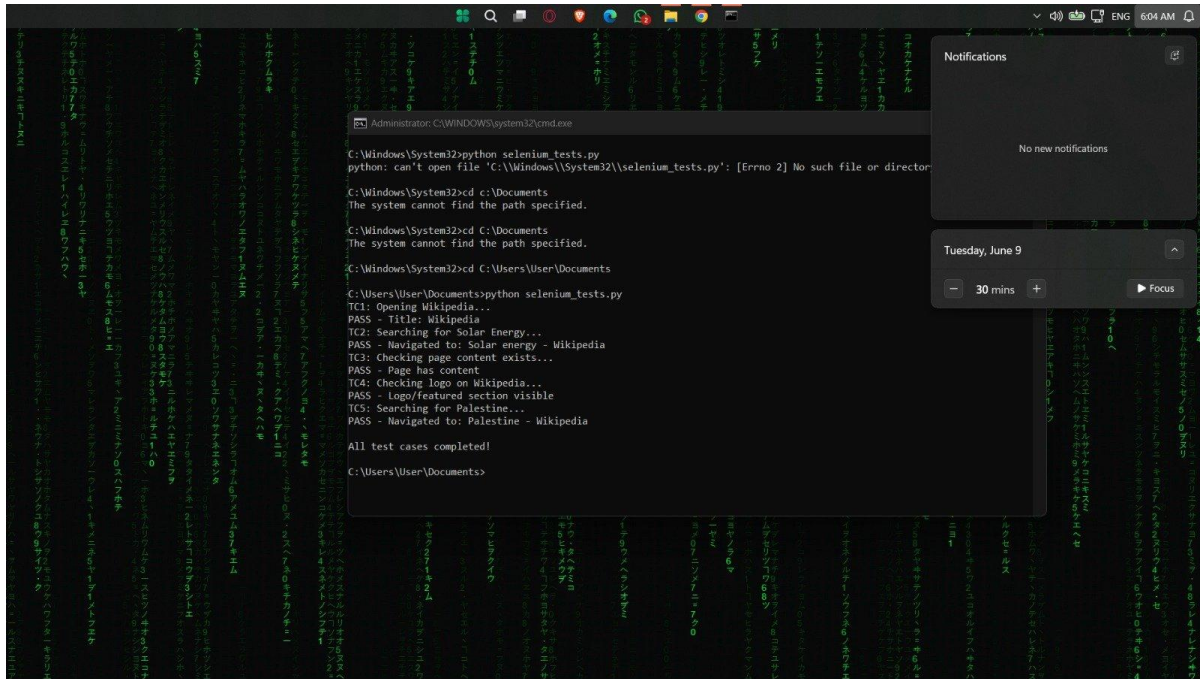


Figure 6: All 5 test cases passed in the Command Prompt — June 9, 2026

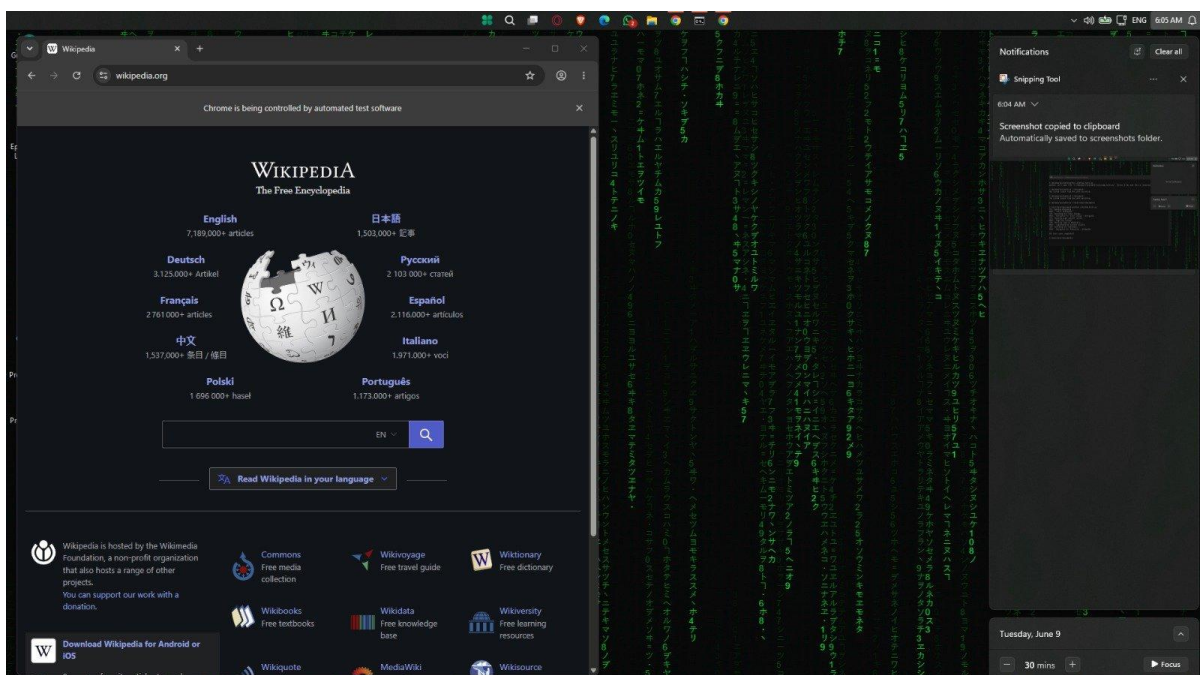


Figure 7: Selenium controlling Chrome — Wikipedia homepage loaded automatically (TC1 & TC4)

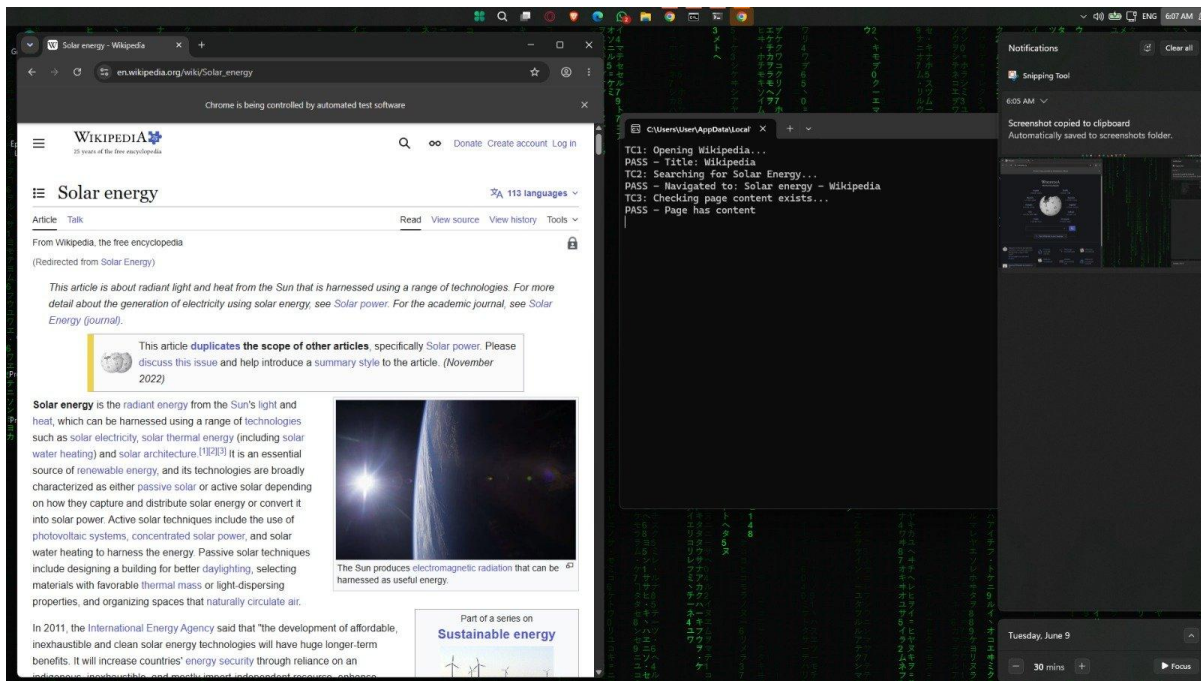


Figure 8: Automated browser navigated to Solar energy article (TC2 & TC3) — June 9, 2026

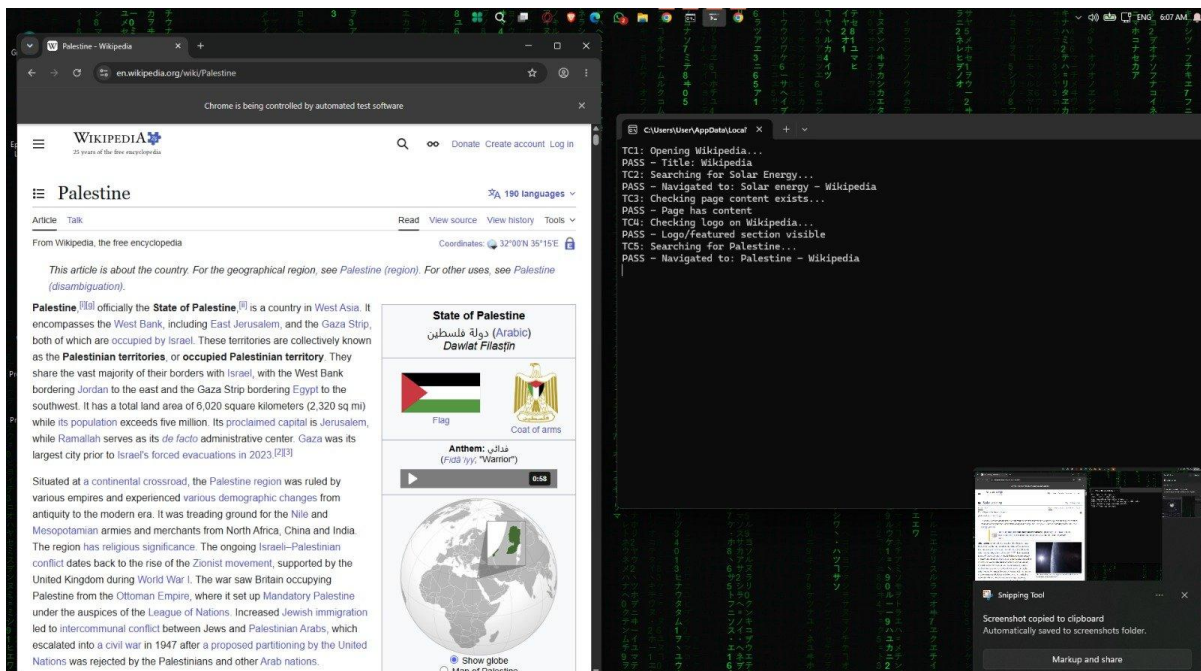


Figure 9: Automated browser navigated to Palestine article (TC5) — June 9, 2026

Conclusion

This project demonstrated the installation and practical use of Selenium, a leading automated testing framework for web applications. Through Part 1, the theoretical foundations of Selenium were explored by watching a professional tutorial video. Through Part 2, the complete toolchain — Python 3.14.5, Selenium 4.44.0, and webdriver-manager was installed and configured on a Windows machine.

Five test cases were designed and executed against the Wikipedia website. All test cases passed successfully, verifying core functionalities including page navigation, search interaction, content verification, and element visibility. The automation process clearly shows how Selenium can simulate real user behavior and validate web application functionality without manual intervention.

This exercise highlights the value of automated regression testing in software development: it saves time, ensures consistency, and provides repeatable test coverage across web applications.

References

- [1] Tech With Tim. (2020). *Python Selenium Tutorial #1 - Web Scraping, Bots & Testing*. YouTube. <https://www.youtube.com/watch?v=Xjv1sY630Uc>
- [2] SeleniumHQ. (2024). *Selenium Documentation*. <https://www.selenium.dev/documentation/>
- [3] Python Software Foundation. (2024). *Python 3.14 Documentation*. <https://docs.python.org/3/>